

F1R3SCORE · WORKING NOTE · SEPTEMBER 2026

Scores as processes

A piece is its parts, put side by side. What passes between them is the music.

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F1R3FLY.IO

WHO THIS DECK IS FOR

Two readers, one notation

For the musician

Every piece is drawn as players, instruments and voices put side by side. Read the lines between them: what crosses a line is how two parts play together, and what does not cross plays freely.

For the AI researcher

Every modelling choice is a choice in the rho calculus with graded where-clauses: notes are communication events, style is a product of graded factors, idioms are spatial and behavioural formulae. The claims are theorems and measurements.

Slides 4–14: pieces as parts in composition. 15–20: one piece, Twinkle, three ways. 21–25: the calculus as a template. From 28: an appendix that opens up the parts.



What a page of notation leaves to the player

Conventional notation is a very good description of one performance.

Where the music is itself a structure of exchange, as in jazz or across many African musics, the grid records only the result.

Which note a line would rather go to

A style is a set of preferences over continuations, not a single written line.

What a phrase is answering

Call and response, trading fours and interlocking parts are structures of exchange.

What a hand can reach

Players follow their hands. Ease of playing shapes every idiom.



COMPONENTS IN COMPOSITION

A piece is its parts, side by side

P | Q | R

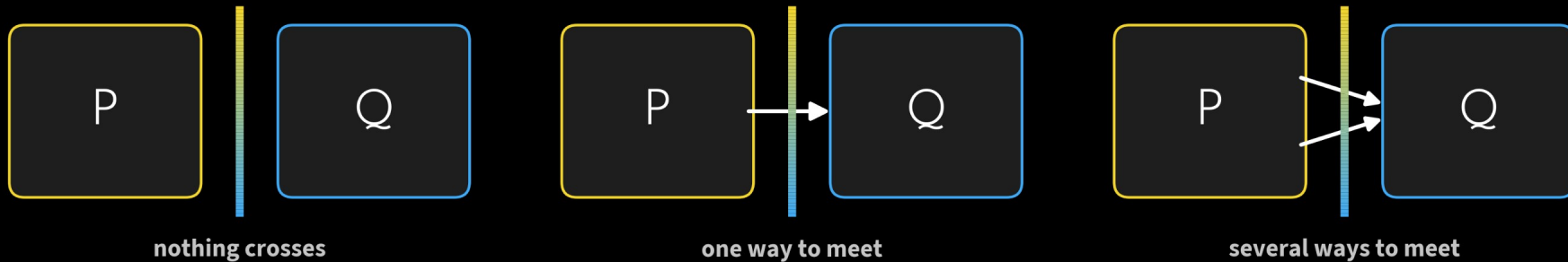
Every score in this deck is written as parts put in parallel, joined by the bar |. Parts can be players, instruments, voices or lines, and the same bar joins all of them.

The bar is a line on the page that can be read. What crosses it is how the parts play together; what does not cross it plays freely.



COMPONENTS IN COMPOSITION

What crosses the line decides what you hear



Harmony

Nothing crosses. The parts are independent, and notes at equal times sound together.

Melody and exchange

One way to meet. Each note hands on to the next, within a part or across the line.

Choice

Several ways to meet. Some candidates must lose, and this is where the weights live.



COMPONENTS IN COMPOSITION

Twinkle, Twinkle, Little Star: Melody | Bass

Melody

piano • 14 notes

C4 C4 G4 G4 A4 A4 G4 · F4 F4 E4 E4 D4 D4 C4

Bass

bass • 8 notes

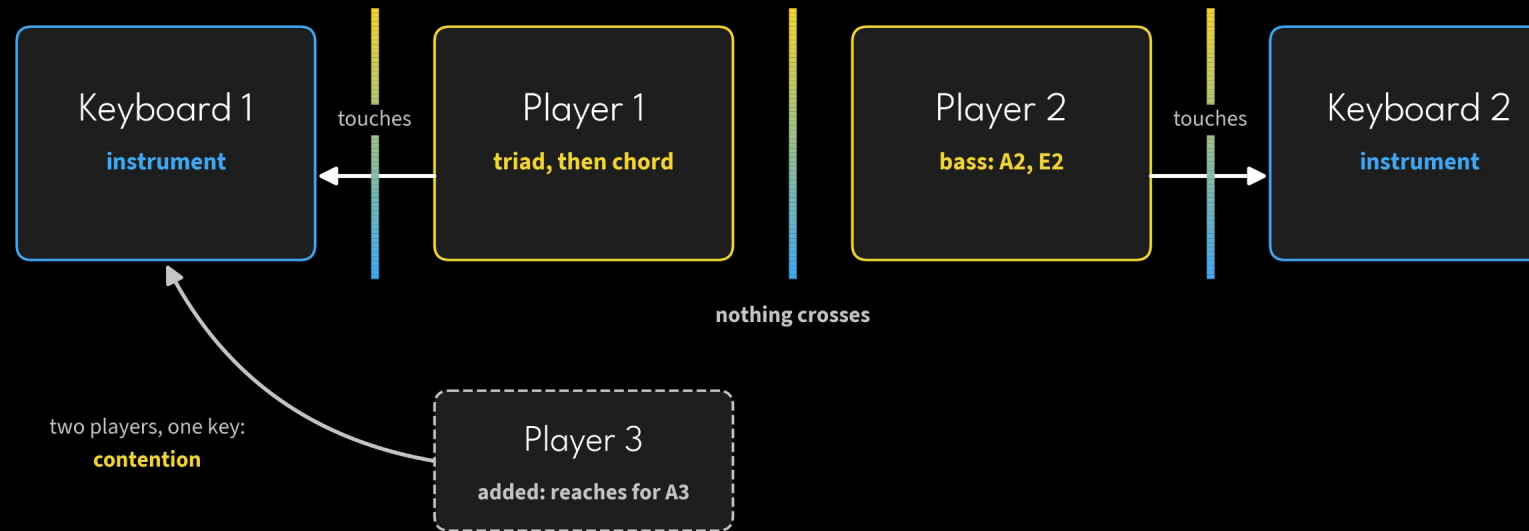
C3 C3 F3 C3 F3 C3 G3 C3

Nothing crosses the line. Melody and bass each keep their own time, and the chords line up without a conductor.



COMPONENTS IN COMPOSITION

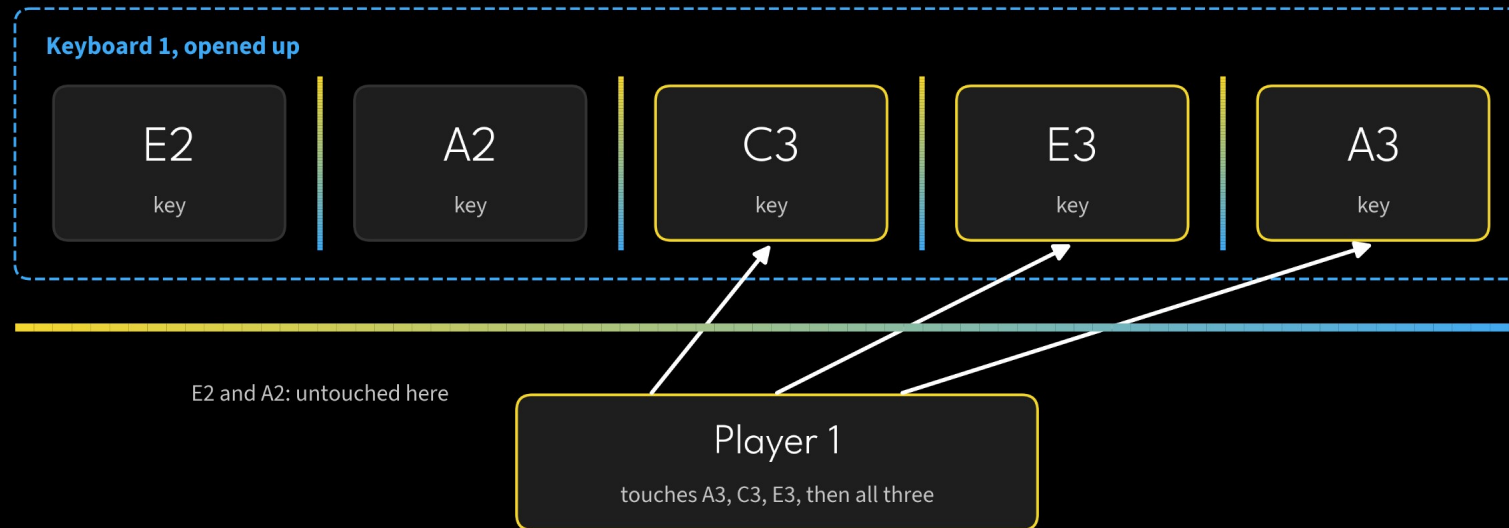
The 4-hands piano: two keyboards, two players



Each player touches only its own keyboard, so nothing crosses between the players and every schedule gives the same performance. Add a third player on the first keyboard and two players reach for one key: exactly two outcomes.

COMPONENTS IN COMPOSITION

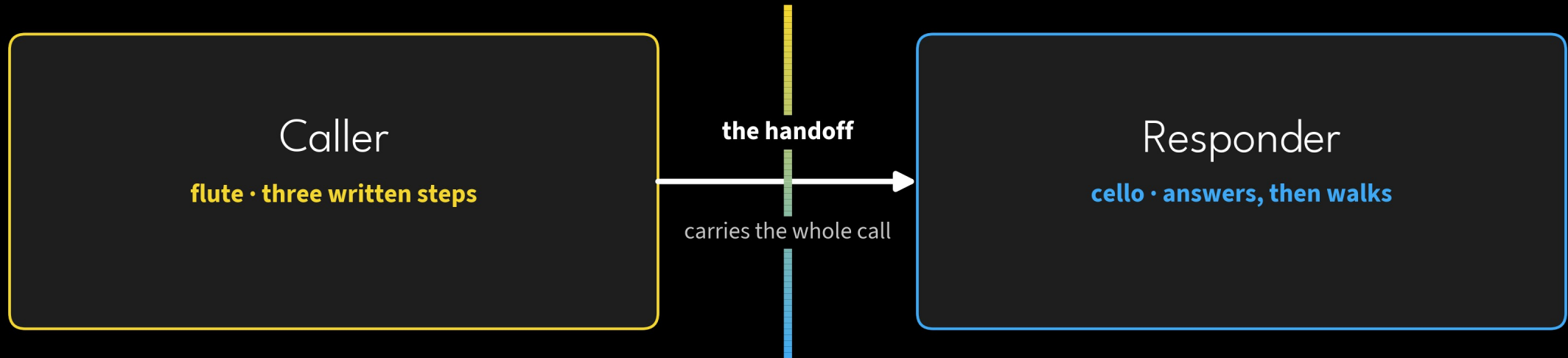
Opening up a part: a keyboard is keys side by side



From outside, Keyboard 1 is one part. Opened up, it is five keys side by side, each a part of its own. Player 1's touches cross into three of them; nothing crosses between keys, so A3, C3 and E3 can sound at once as a chord.

COMPONENTS IN COMPOSITION

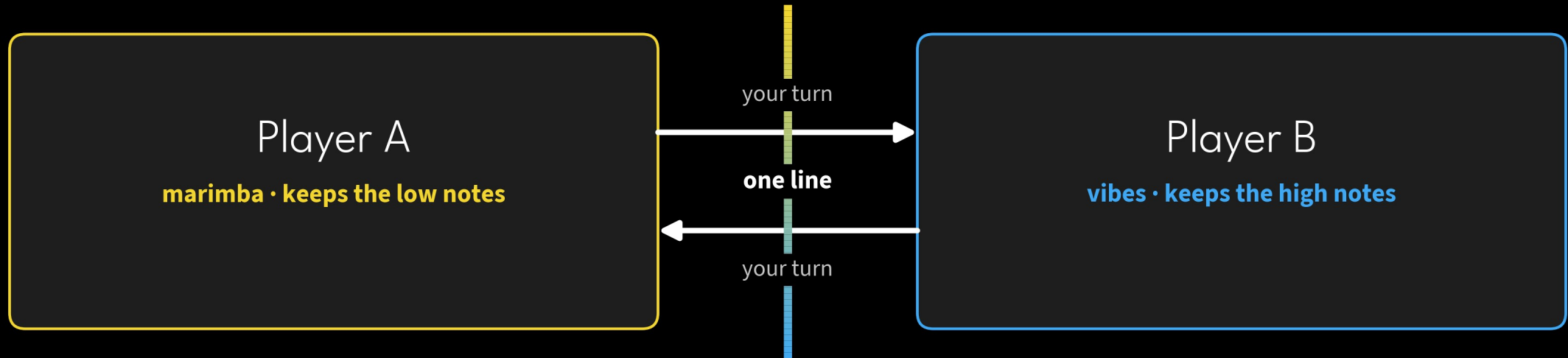
Call and response: Caller | Responder



One thing crosses: the handoff. It carries the caller's whole phrase, so the responder can answer it, here a fifth higher when the call ends on G4.

COMPONENTS IN COMPOSITION

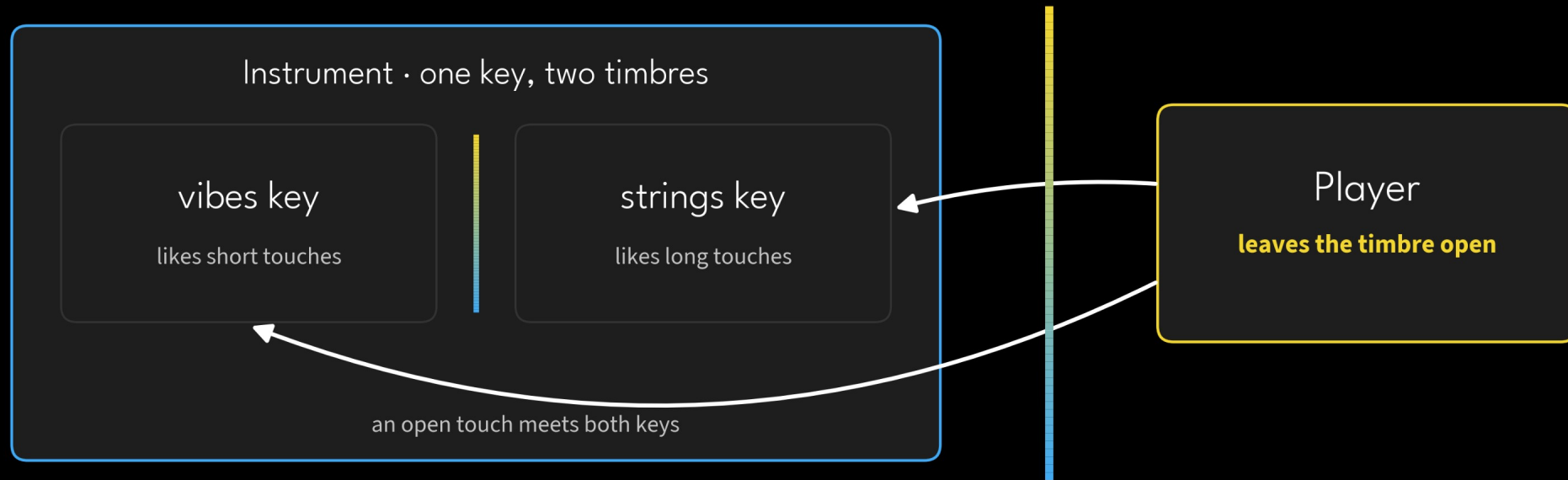
Hocket: two players, one line



The line crosses back and forth, one note at a time: each player's note asks the other for the next. The marimba keeps the low notes, the vibes the high ones, and one set of preferences shapes the whole line.

COMPONENTS IN COMPOSITION

A chimeric instrument: articulation picks the sound



A player who leaves the timbre open touches both keys at once, and they compete. Detached touches are heard mostly on vibes, sustained ones mostly on strings. Name the timbre and it is an ordinary synthesizer.

COMPONENTS IN COMPOSITION

Two voices, each with its own style

Voice A

piano

country × stepwise × rhythm

Voice B

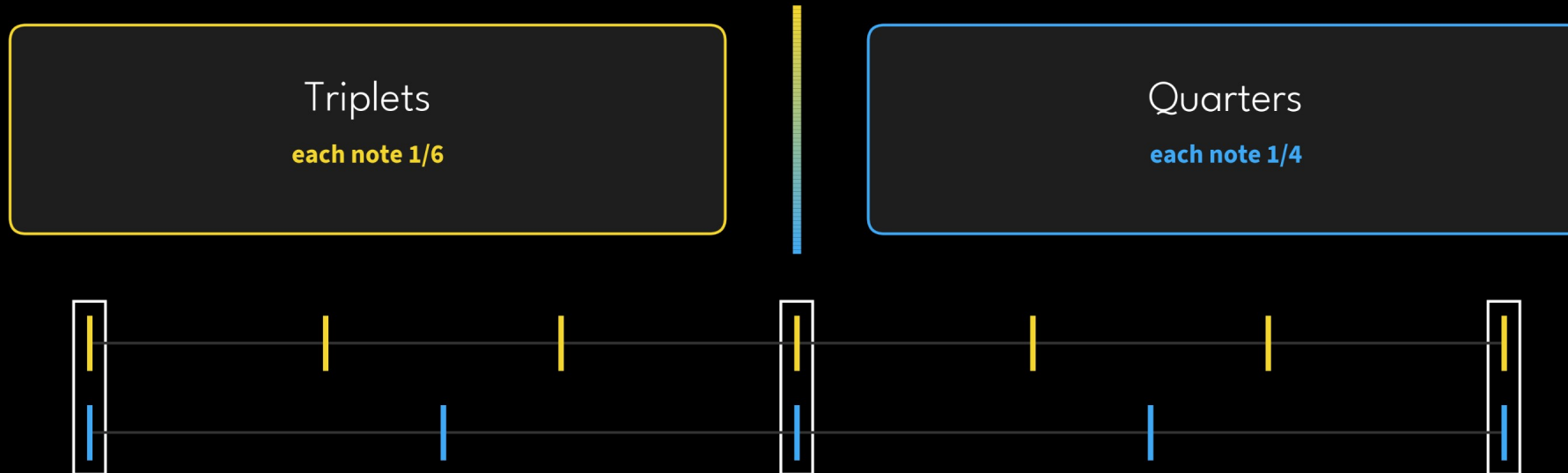
bass

walking × slow

Style lives inside a part: each voice multiplies its own preferences. Between the parts nothing crosses, so the voices harmonize freely and each keeps its own clock.

COMPONENTS IN COMPOSITION

Three against two



Two independent parts, one in triplets and one in quarters. Nothing crosses, yet their onsets meet at every half note, whatever pitches are chosen.

COMPONENTS IN COMPOSITION

Reading the lines

Piece	Parts	What crosses	What is heard
Twinkle	Melody, Bass	nothing	a melody over a bass; the chords line up
Two voices	Voice A, Voice B	nothing	free harmony in two styles
Three against two	Triplets, Quarters	nothing	two pulses meeting every half note
4-hands piano	2 keyboards, 2 players	touches, each to its own keyboard	an arpeggio and a chord over a bass
Call and response	Caller, Responder	one handoff, carrying the call	an answer that hears the question
Hocket	Player A, Player B	the line, note by note	one line shared by two instruments
Chimera	Instrument, Player	an open touch, to two keys at once	a timbre that follows articulation

DEEPER DIVE

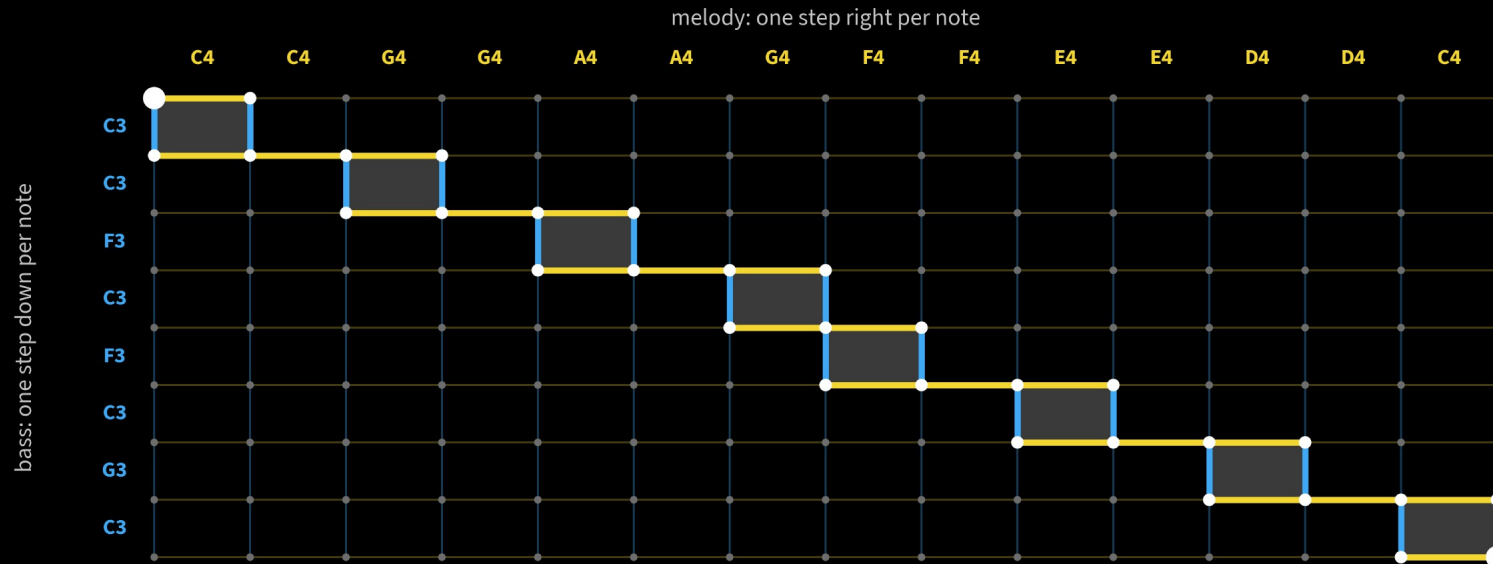
One piece, three views

Twinkle, Twinkle, Little Star as a graph of everything its two parts can do, as the shorthand that writes it down, and as the traditional score it sounds like.



DEEPER DIVE · TWINKLE THREE WAYS

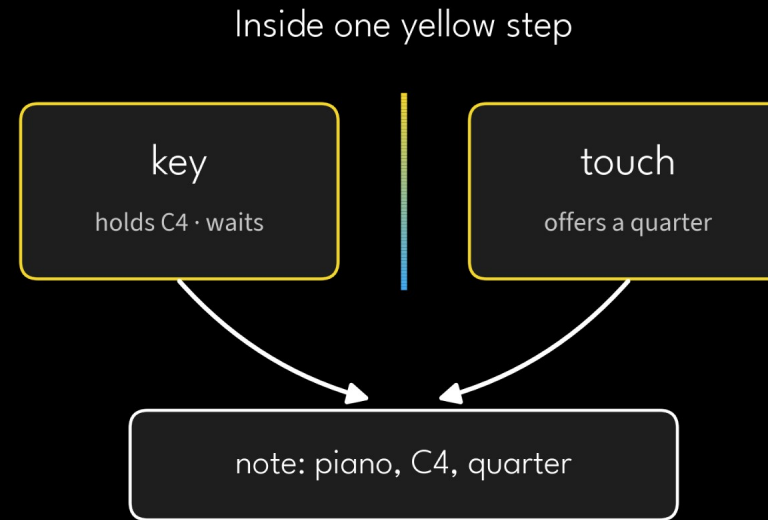
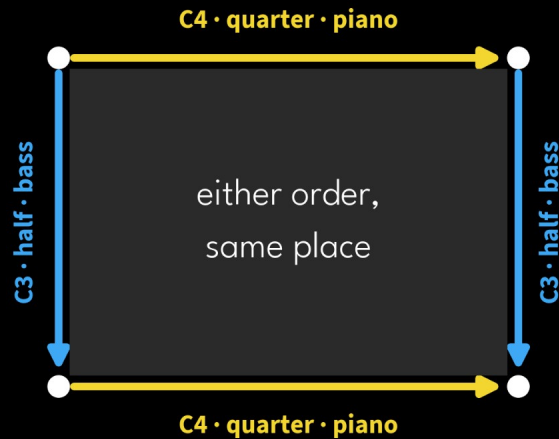
View 1 · The reduction graph



Start at top left with Melody | Bass. Each yellow step plays a melody note, each blue step a bass note; every path ends at bottom right with both parts done. Played in time, a run stays in the lit corridor, and its eight shaded squares are the eight chords.



Zooming in: independence and interaction



Left: a melody step and a bass step commute. That is independence, and it is why harmony needs no conductor. Right: inside a step, a key and a touch in one part meet; the note is their meeting, and it brings the next key and touch forward.

View 2 · The shorthand

```
score Twinkle
import std
pitches { r, C3, F3, G3, C4, D4, E4, F4, G4, A4 }
durations { q = 1/4, h = 1/2 }
timbres { piano = gm(0) on 1, bass = gm(32) on 2 }

play line(base "Lm", piano)
    [ C4 q, C4 q, G4 q, G4 q, A4 q, A4 q, G4 h,
      F4 q, F4 q, E4 q, E4 q, D4 q, D4 q, C4 h ]
| line(base "Lb", bass)
    [ C3 h, C3 h, F3 h, C3 h, F3 h, C3 h, G3 h, C3 h ]
```

C4 q

One yellow step: a key holding C4 meets a touch offering a quarter.

C3 h

One blue step down.

|

The whole grid: two lines side by side, free to go in any order.

No weights

A fixed piece: no where-clauses, one way to meet at every step.



View 3 • The traditional score

shared onsets 1 2 3 4 5 6 7 8

The image displays a musical score for 'Twinkle, Twinkle, Little Star' in 4/4 time. The score is written on two staves: a treble staff for the melody and a bass staff for the bass line. The melody is written in yellow, and the bass line is written in blue. The score is divided into measures by vertical bar lines. Above the staff, the first measure is labeled 'shared onsets', and the subsequent measures are labeled with numbers 1 through 8. Below the staff, Roman numerals I, IV, and V are placed under the corresponding measures. The melody and bass line are synchronized, with the melody starting on a half note and the bass line starting on a half note. The melody consists of a sequence of eighth notes, and the bass line consists of a sequence of half notes. The score ends with a double bar line.

The staff keeps the one timed performance that every run through the corridor agrees on. Its eight numbered onsets are the graph's eight shaded squares: the chords sound where melody and bass meet in time, though nothing passes between them.

Three views of one process

View	What it shows	What it leaves out
Reduction graph	every order the parts can play in; independent notes commute	when each note sounds
Shorthand	the parts and the bar between them; each note as a meeting	the orders the bar permits
Traditional score	the one timed performance, and its chords	that melody and bass never communicate

Read in this order, the traditional score is the last and flattest view of a process that the first two views show whole.

Why this is worth an AI researcher's time

A generative model of music in which every design choice is a choice in the rho calculus with graded where-clauses, and every claimed behaviour is a theorem or a measurement.

Concrete

Alphabets are finite, weights are exact rationals, and a reference simulator and a Rust player implement the same semantics.

Compositional

Style, physicality and idiom are separate graded factors with a proven law for their product.

Constrained and probabilistic

Hard rules and soft preferences are different sorts of formula, and they provably do not interfere.



Reading the calculus as a template

in the calculus	in the score	the general pattern
COMM on matching subjects, wildcards allowed	a note: pitch meets duration	each event a typed interaction
unlabelled reduction; records read by v	the note read off each record	provenance first, labels derived
polarity: receipt holds pitch, message duration	keys to instruments, touches to players	agent and embodiment split by type
weights normalised per contention set	which continuation is played	weighted, normalised rule choice
$\otimes$ of graded factors	style $\times$ register $\times$ physicality	product of experts; 0 is a veto
crisp $[\beta]$ beside graded ψ	rule beside tendency	support apart from likelihood
spatial atoms over the quoted location	motifs read from the past	memory as a structured name
$\langle K \rangle$ modalities with a vX viability	an idiom kept in company	shielded generation, no deadlock
reflection: $@P$ and $*x$	unbounded voices, silently	recursion by reflection alone

What is proved rather than hoped for

Law of a voice

Each fitting is drawn from the normalised product of the clause's factors, conditioned on the past.

Factorised resolution

Factors over disjoint aspects are heard independently and can use separate chance sources.

Support and grading

Positive weights never change which traces are admissible; crisp factors only shrink the support.

Rules without deadlock

With a viability greatest fixed point, a voice never breaks the rule and never falls silent.

Free harmony

Independent voices have product law and the same performance under every interleaving.

Two-sided fitting

A single note cannot choose from both sides at once; one datum is fixed a note early.



The ideal defines; the checkable decides

The viability formula is a greatest fixed point in the ideal logic, quantifying over the contexts the rest of the score supplies.

When a rule reads a bounded window of the past and the alphabets are finite, the relevant state is the window, the held pitch and the recorded duration. That is a finite graph, on which the fixed point is computed by iteration.

This is the ideal-versus-checkable discipline of the rho-life notes, applied to a working generator.

OPEN THREADS

Learning styles

Paper II's gradient in the fibre can fit a style factor to a corpus; the composer's formulae say which features the learner may use.

Complex grading

Over $\mathbb{C}$, two phrasings reaching one configuration would interfere; a voice already has what the quantum fragment needs.

Tempo

A second clause, valued in a second resolution algebra, that scales durations as continuations are stamped.



Checked by simulation, played by a Rust player

skeinsim.py · reference simulator

LAW OF A VOICE

20,000 notes; largest deviation from the pitch matrix 0.031 and from the duration matrix 0.009.

RECURSION BY REFLECTION

identical to the step semantics over 3,000 notes; exactly two null notes per note, all silent.

THE UNPLAYED IS INERT

$79 = 5 \cdot 16 - 1$ inert messages left per note over 400 notes; none ever joins a candidate.

F1R3Score · the player

FORMATS

a table, JSON, a MIDI file or live MIDI, and a trace of records that replays exactly and plays back without the score

CORE

exact rationals, wildcard decorations, one playback function on records, hash-consed terms, maximal matchings

TESTS

70, including T0–T14 with the 4-hands piano and the chimera; a differential with skeinsim.py agrees record for record



GETTING STARTED

Where the work lives

F1R3FLY-IO/PUBLICATIONS · F1R3SCORE

Scores as Processes (score-calculus), the player specification, and skeinsim.py

F1R3FLY-IO/F1R3SCORE

the Rust player (check, expand, play, replay, render) and its guide to building, deploying and using it, docs/f1r3score-guide.pdf

F1R3FLY-IO/PUBLICATIONS · FUZZYWARE

the graded where-clause papers: Paper I (quantum) and Paper II (over the reals)

F1R3FLY-IO/F1R3GAMES · F1R3SKEIN

the virtual instrument the idea grew out of

TRY IT

```
cargo build --release
f1r3score play  examples/four_hands.score \
    --midi fourhands.mid
f1r3score play  examples/chimera.score \
    --midi chimera.mid
f1r3score play  examples/hocket.score \
    --notes 16 --stream vibes=human
```

The last command lets you improvise the vibes part yourself while the score plays the marimba.



A score is its parts side by side. A performance is one path through what they can do together.

Scores as Processes · F1R3FLY.io working note, September 2026



F1R3FLY.IO

APPENDIX

Inside the components

The main slides show pieces as parts in composition. What follows opens the parts: how a note is a meeting, how a voice chooses, how style and physicality are weighted, how idioms become formulae, and how instruments are built.



Three things the score writes down explicitly

1

Who plays with whom

Players are parts of the process.
Handing a line to another player, or
alternating the notes of one line
between two players, is a
communication.

2

How likely each continuation is

Every choice weighs its alternatives.
Factors for style, register and
instrument multiply; the normalised
product is what plays.

3

What the music must or should do

Formulae read the recent past (motifs)
and look ahead in the company of the
other players (idioms). Rules forbid;
weights encourage.

A score written this way is a distribution over performances.

A note is where a pitch meets a duration

THE KEY · A CHANNEL

```
for (_ ← <@L, piano, C4>)  
{ ...next... }
```

Holds a pitch and waits.



C4 · quarter

one note, emitted



THE TOUCH · A CO-CHANNEL

```
<@L, piano, q>!(...)
```

Offers a duration.

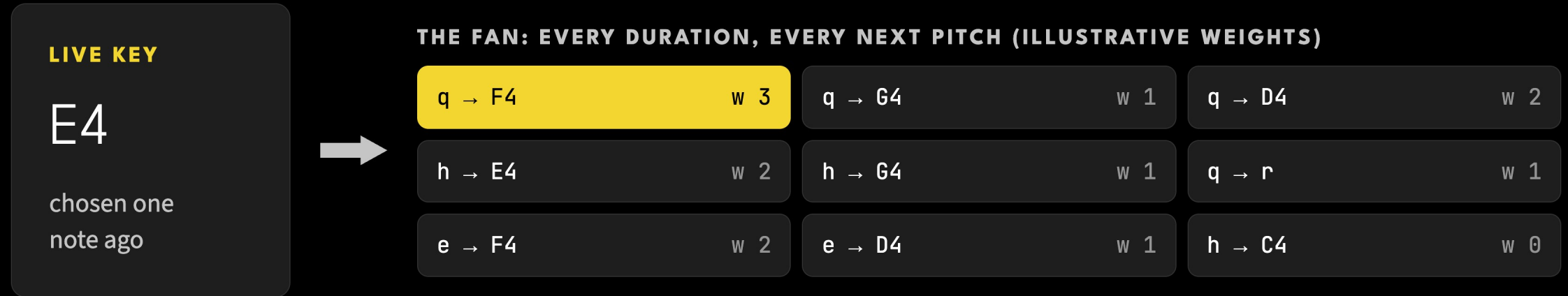
They meet when their subjects match: both at **@L**, timbres that meet (either may be left open, **_**), a pitch facing a duration.

Every meeting leaves a record, and the note is read off the record. There are no silent steps anywhere in the calculus.

A rest is a pitch like any other. A rest of length zero is how the machinery of a score stays inaudible.



A voice: a key, a fan, and its own past



The key already holds its pitch. The fitting chooses how long it lasts and which pitch comes next.

The chosen record becomes the next location, so a voice's location is its past, most recent note outermost.

Everything not chosen is left where no receipt will ever look. The unplayed is inert.

Style and physicality are separate factors

```
factor stepwise = table step { -2..2: 3, _: 3/10 }
factor country  = machine pitch { C4 → D4 @ 2,
                                   C4 → E4 @ 3, ... }
factor rhythm   = machine dur { q → q @ 3, q → e @ 2, ... }

play Voice(piano, country * stepwise * rhythm,
           C5, q, "A") #A
  | Voice(bass, walking * slow,
         C4, h, "B") #B
```

Product *

All preferences at once. A continuation is likely only if every factor accepts it, and any factor can veto with 0.

Sum +

A blend of styles. Scaling one summand sets the mix.

Interval factors

Weights that depend only on the step make the line a sum of intervals, so combining them convolves interval laws.

Only ratios are heard: a composer writes weights on whatever scale is convenient.

One style, two instruments

A fixed style over twenty-one pitches is multiplied first by a stepwise instrument (steps of at most two degrees weighted 3, all else 0.3), then by a leaping one (intervals of four or seven degrees weighted 3, all else 0.3).

measured over 20,000 notes each	stepwise instrument	leaping instrument
share of steps of at most two degrees	0.582	0.135
share of leaps of four or seven degrees	0.051	0.378
mean interval, in degrees	4.91	6.97

In both cases the played transitions match the normalised product, and no transition the product forbids ever occurs. Imagined physicalities, such as a hand spanning three octaves or strings tuned in thirds, are one-line changes to the score.

Motifs and rules read the voice's past

ENCOURAGE · A GRADED FACTOR

Complete the motif 0, 2, 4

```
[ last[0, 2] && carry(4) ] * 20 + 1
```

The head 0, 2 is completed by 4 at a rate that rises from 0.171 to 0.775 over 10,000 notes, against analytic values of 0.154 and 0.792.

ENFORCE · A CRISP FACTOR

No pitch three times in a row

Unconstrained, a line of 10,000 notes contains 1,303 threefold repeated pitches. With a crisp factor forbidding a pitch equal to the last two, it contains none, and the voice never falls silent.

Transformations of a motif are formulae too: an answer transposed by a fixed interval, an inversion, or an echo of the rhythm with new pitches.



Improvisation is a trace through the score

THE CRISP LAYER

What may be played: the tune, the changes, the rules of the idiom. A crisp factor can only shrink the set of admissible traces.

THE GRADED LAYER

What the idiom would rather play. Changing positive weights changes how likely a trace is, never whether it is admissible.

Algorithmic

samples a trace with the probabilities the grading gives.

Human improviser

chooses freely among the candidates of positive weight.

Mixed

an accompanist driven by the score follows a human soloist.

With a viability factor, a performer who stays within the score can never be left with nothing admissible to play.



Exchange between players is written directly

Call and response

A caller hands the line to a responder. The handoff is itself a note, and it carries the caller's whole history, so the response can read the call: answer a fifth higher, or echo its rhythm.

Hocket and interlocking

Two players share one line, each serving the other's next note. The line's factors see every note while each player adds their own preferences. They may play one instrument, as on an amadinda, or two: the line can pass from marimba to vibes note by note.

Cross-rhythm

Three against two is two independent voices, one using a triplet value of $1/6$ and one a value of $1/4$. Onsets coincide at every half note whatever pitches are chosen, and each part keeps its own pulse.



Free harmony, coupled voicing

INDEPENDENT VOICES

Product law, same performance under any schedule

Five schedulers over 3,000 resolutions produced identical performances, with 216 simultaneous strikes. Nothing coordinates the voices; each keeps its own clock.

CHORDS AS JOINS

One receipt that weighs a whole voicing

A chord receipt waits for several durations at once and sees every tone of the next chord together. That is where vertical physicality lives: closed stacked thirds under a pianist's hand, open fifths on strings tuned in fifths.

A hand is not a set of independent fingers. The calculus keeps harmony and voicing apart, and the composer chooses which relation holds between which parts.



Keys, touches and synchronous output

A key belongs to the instrument

It holds a pitch and re-arms itself after each strike, so two touches aimed at one key queue rather than overlap.

A touch belongs to the player

It offers a duration, and may leave the timbre open for the instrument to supply.

; waits for the note to end

A chord is touches in parallel; an arpeggio is the same touches joined by ;.

Onset	Keyboard 1	Keyboard 2
0	A3 q	A2 h
1/4	C3 q	
1/2	E3 q	E2 h
3/4	A3 h, C3 h, E3 h	

Inside the chimera: several keys at one location

Each key location holds one re-arming key per timbre. A touch that names a timbre meets only that key; an open touch meets all of them, and the keys' clauses, which see the touch's duration and payload, decide which sounds.

Weights: vibes 9, 1, 1 and strings 1, 1, 9 for eighth, quarter and half. The losers are never consumed, so the chimera stays whole, and a second open touch during a note can sound in the other timbre.

Player's touches	Heard on vibes
detached eighths, timbre open	about 0.9
quarters, timbre open	about 0.5
sustained halves, timbre open	about 0.1
any, timbre named vibes	exactly 1

Where the idea came from: F1R3Skein

F1R3Skein is a virtual instrument that fits a pitch stream and a duration stream into notes: first from the digits of transcendental constants, later from weighted state machines the player draws.

Two streams meeting at a note suggested channels and co-channels. The aims differ: the instrument is for a performer, and the calculus is a notation for a composer. The instrument is one score among many.

```
f1r3score play t4_embedding.score \  
  --notes 16 --gc freshness \  
  --stream A.pitch=spigot:pi/16@1 \  
  --stream A.dur=spigot:e/5
```

The pitches are the hexadecimal digits of π and the durations the base-5 digits of e . Over 400 notes the streams equal the digit streams, one digit per draw.

